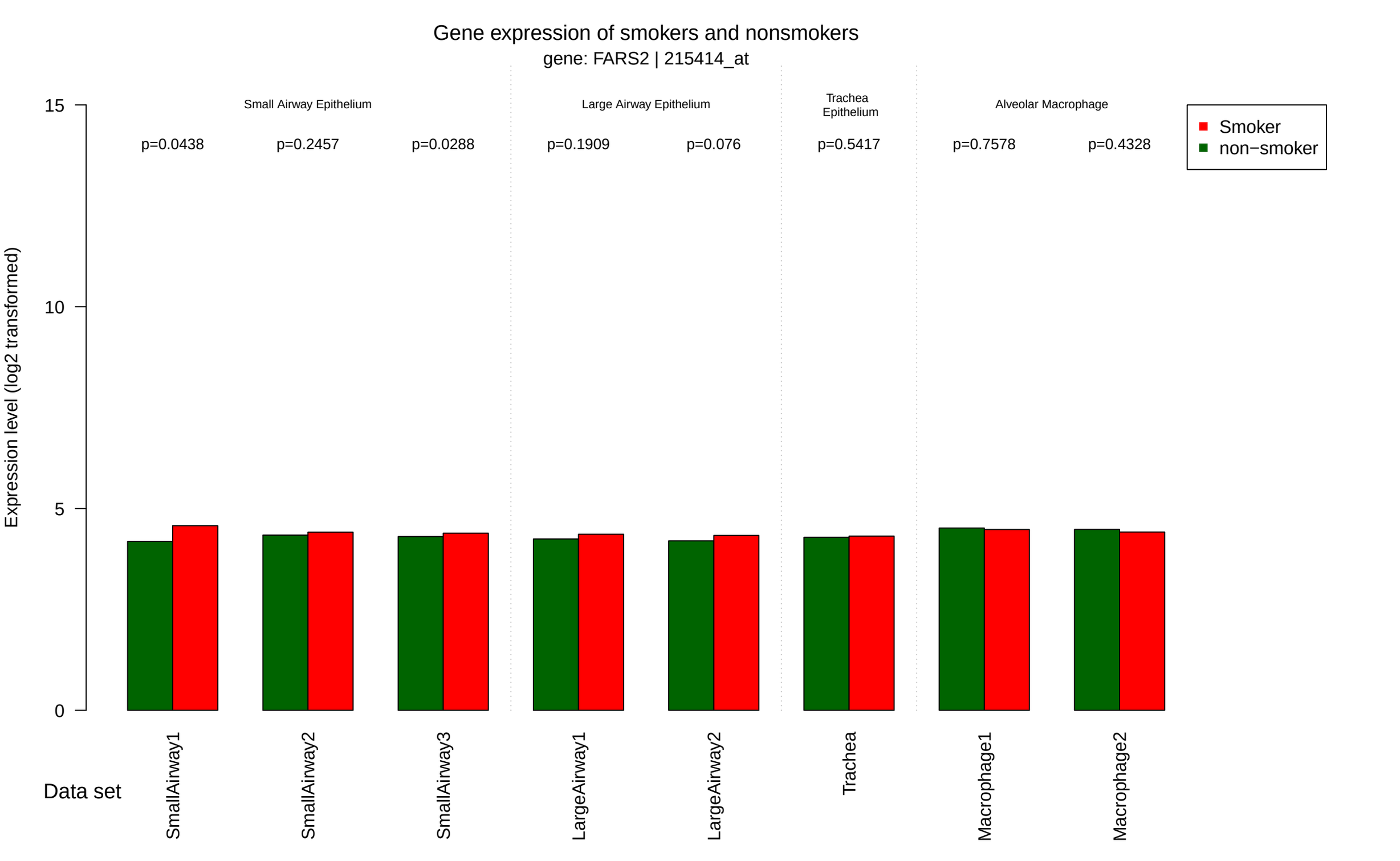
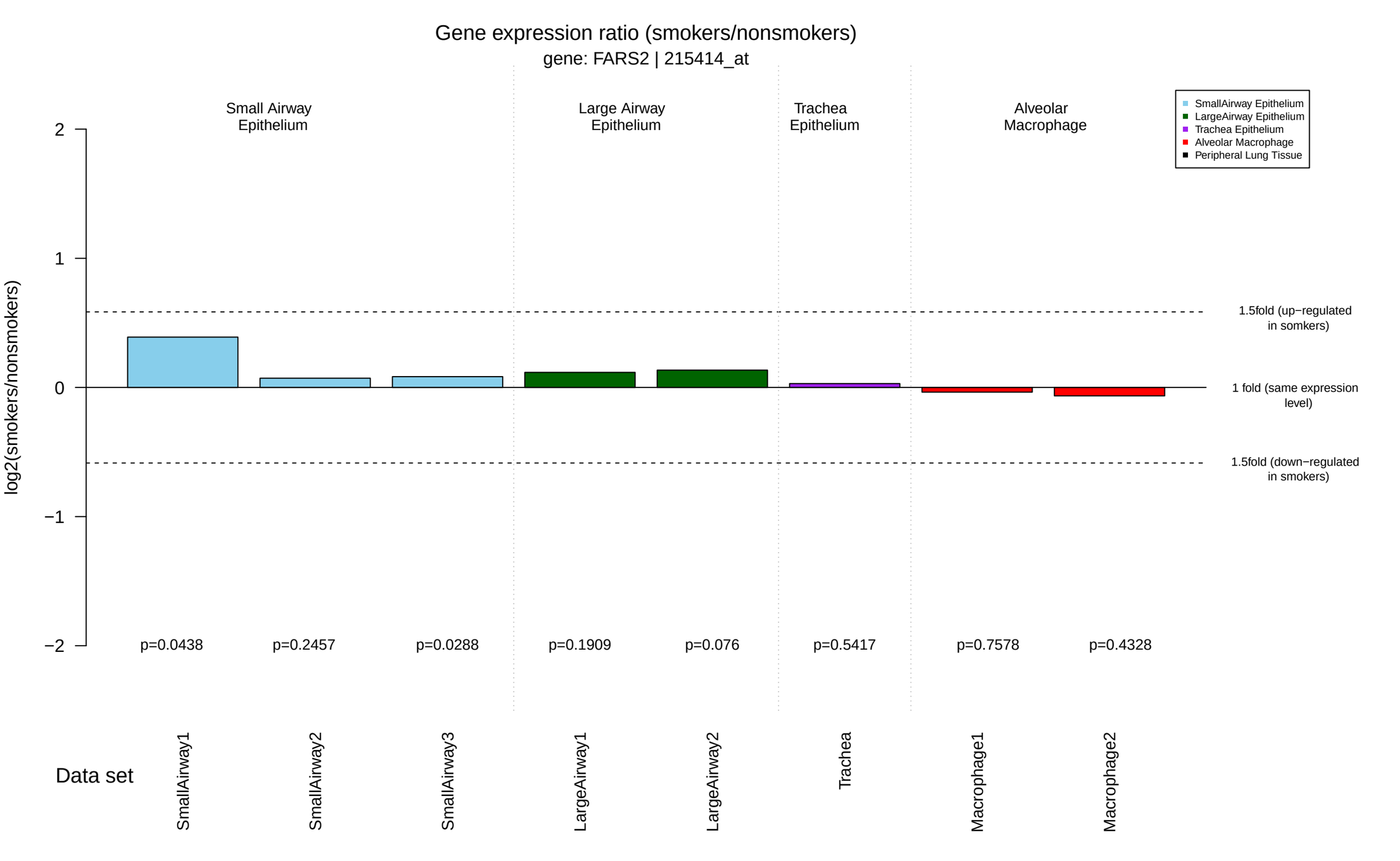
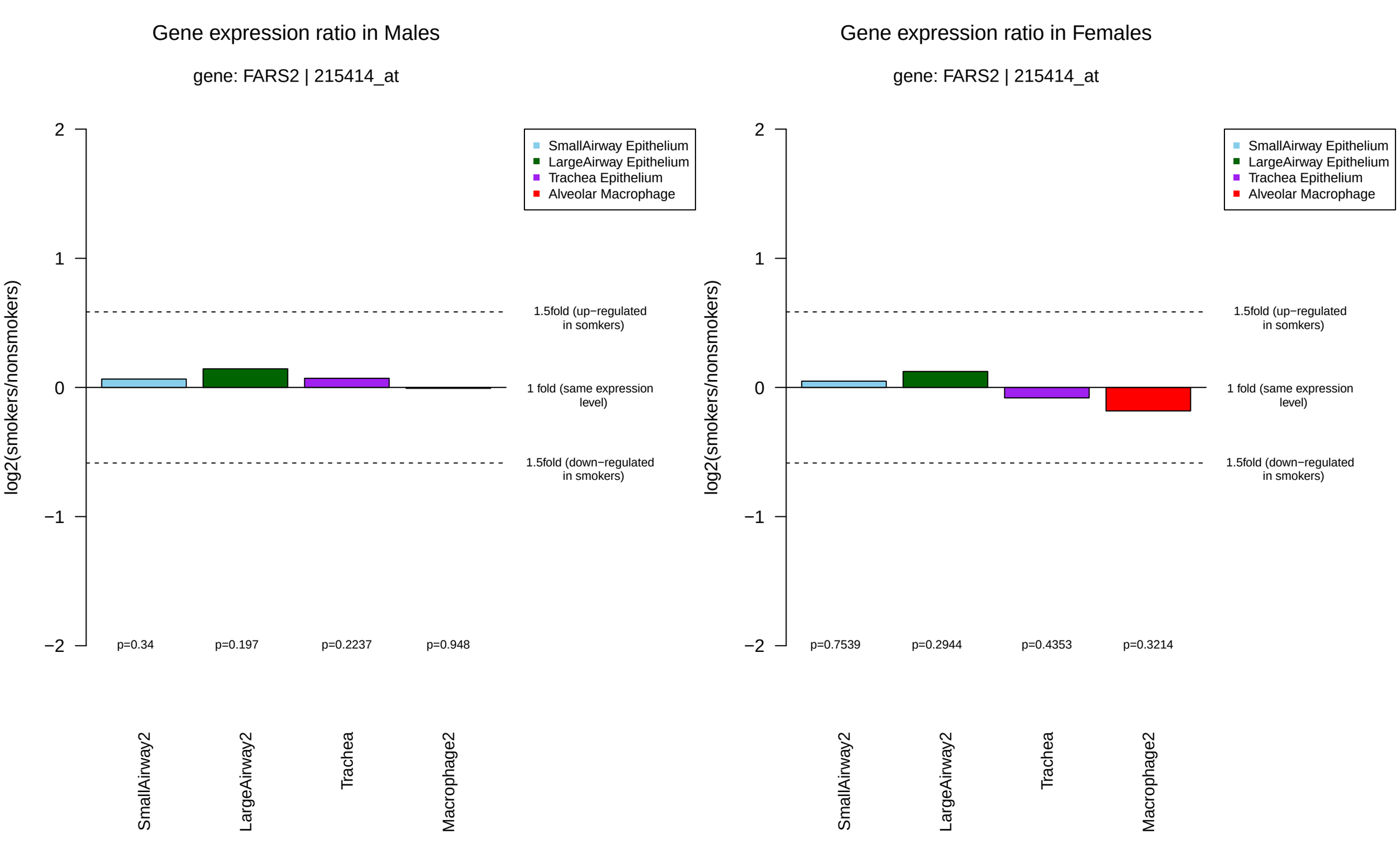
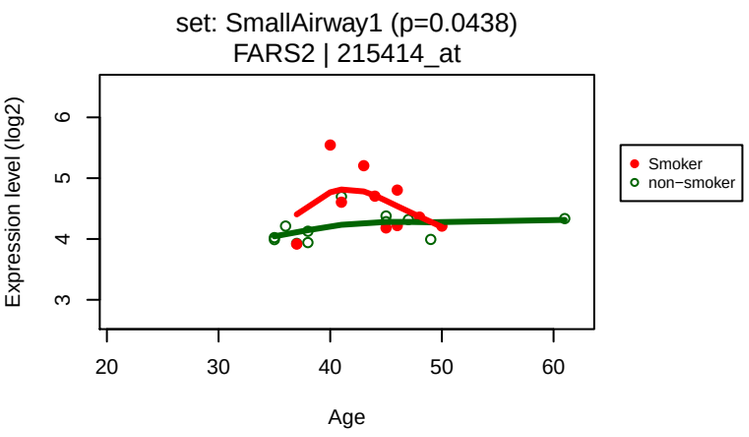
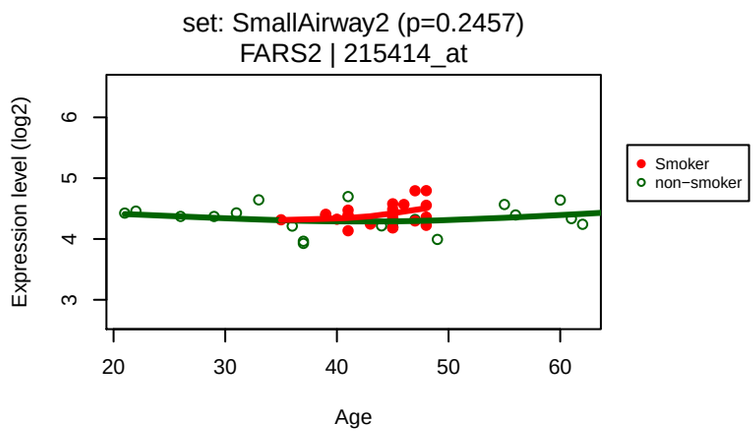
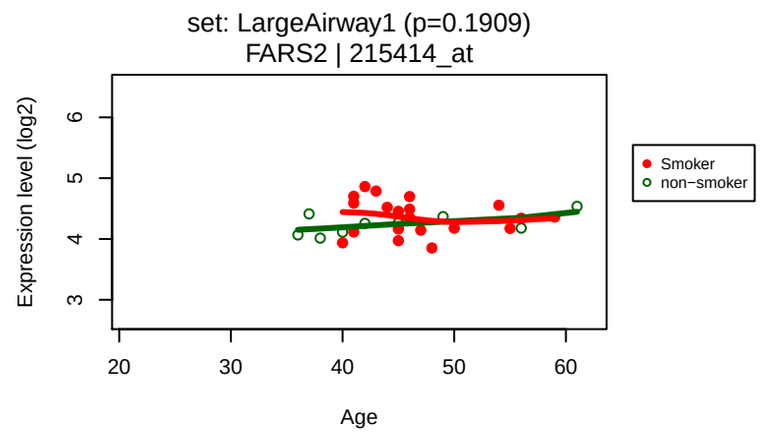

Gene expression of smokers and nonsmokers
gene: FARS2 | 215414_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0438
p=0.2457
p=0.0288
p=0.1909
p=0.076
p=0.5417
p=0.7578
p=0.4328
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

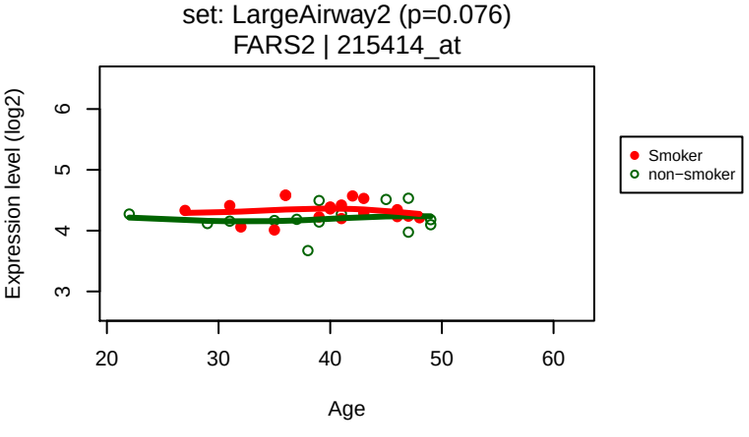

Gene expression ratio (smokers/nonsmokers)
gene: FARS2 | 215414_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0438
p=0.2457
p=0.0288
p=0.1909
p=0.076
p=0.5417
p=0.7578
p=0.4328
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

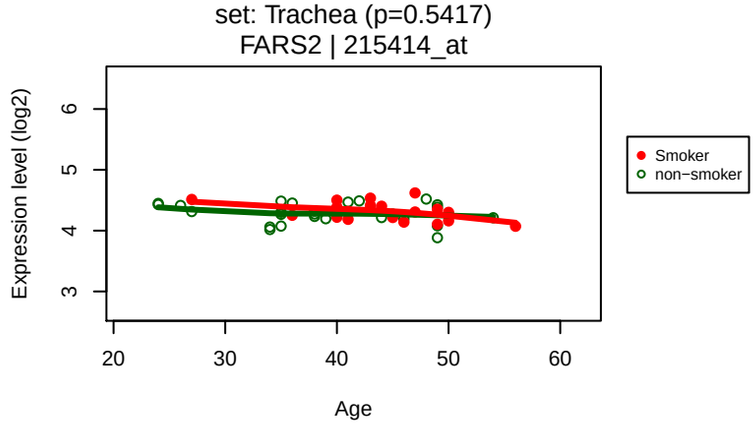

Gene expression ratio in Males
gene: FARS2 | 215414_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.34
p=0.197
p=0.2237
p=0.948
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FARS2 | 215414_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7539
p=0.2944
p=0.4353
p=0.3214
SmallAirway2
LargeAirway2
Trachea
Macrophage2

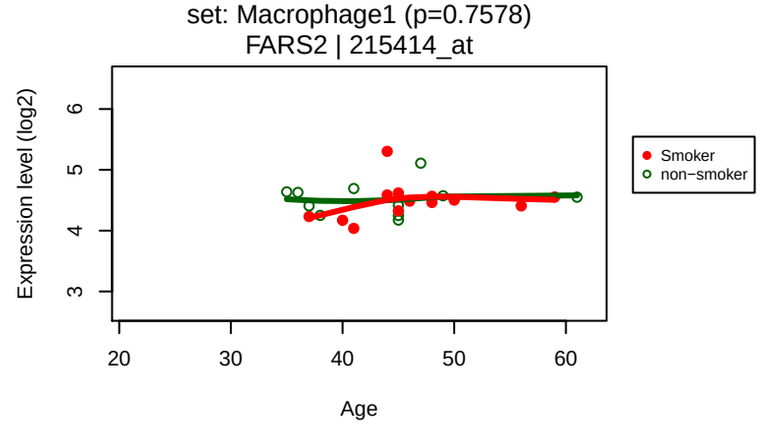

set: SmallAirway1 (p=0.0438)
FARS2 | 215414_at
Expression level (log2)
Age
Smoker
non−smoker

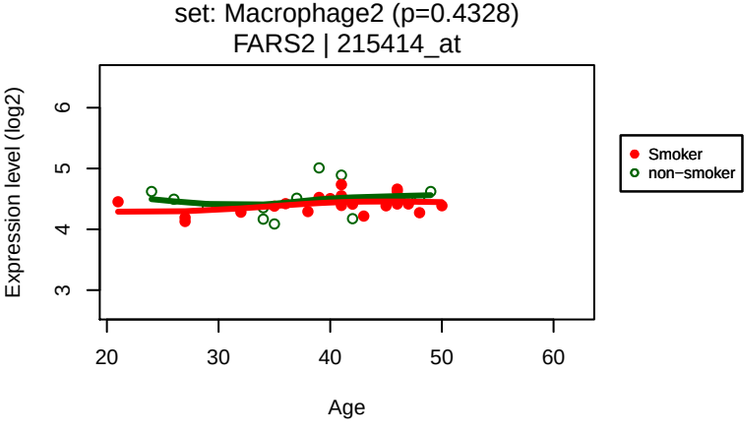

set: SmallAirway2 (p=0.2457)
FARS2 | 215414_at
Expression level (log2)
Age
Smoker
non−smoker

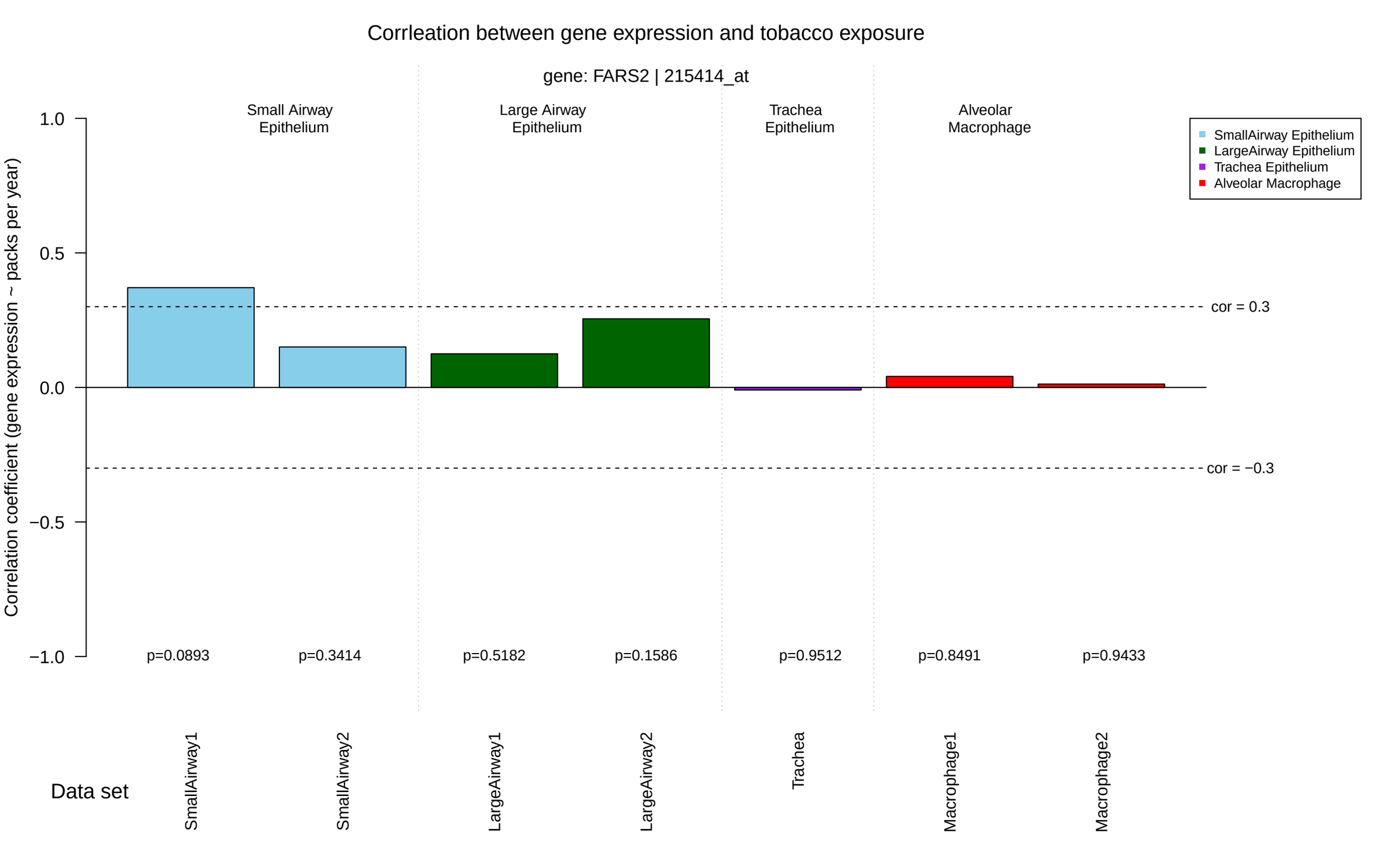

set: LargeAirway1 (p=0.1909)
FARS2 | 215414_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.076)
FARS2 | 215414_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5417)
FARS2 | 215414_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7578)
FARS2 | 215414_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4328)
FARS2 | 215414_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FARS2 | 215414_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0893
p=0.3414
p=0.5182
p=0.1586
p=0.9512
p=0.8491
p=0.9433
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2